Light

Data Representation — Media (Text, Images, Sound)

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Edexcel GCSE Computer Science — **Data Representation: Images and Sound (1CP2/01 Topic 2: Data)**. This topic covers how computers represent bitmap images and sound in binary, including the key concepts of resolution, colour depth, sample rate, and bit depth, plus file size calculations.

Exam questions typically ask you to **describe** how images or sound are represented, **explain** the effect of changing resolution or sample rate, and **calculate** file sizes. The 2024 Paper 1 included questions on image representation and file size calculations. Expect questions worth 2—6 marks, often requiring you to construct expressions and perform calculations without a calculator.

Why data representation matters

Everything stored on a computer must be represented in binary (0s and 1s). Understanding how images and sound are converted to binary is essential because:

- **File sizes:** Knowing how data is stored helps you calculate and predict file sizes
- **Quality vs storage:** Higher quality means larger files — understanding the trade-offs is crucial
- **Exam calculations:** File size questions appear regularly and require you to apply formulas correctly

- **Real-world relevance:** Streaming services, cameras, and music players all use these concepts
- **Paper 1 Topic 2:** This content typically appears as calculation questions worth 3—6 marks

What you must know

This checklist covers the key exam-visible concepts from Edexcel 1CP2 Topic 2: Data. You should be able to explain each point and apply it to exam scenarios.

- **Describe** how bitmap images are represented in binary using pixels, resolution, and colour depth
- **Explain** the effect of changing resolution and colour depth on image quality and file size
- **Construct expressions** to calculate image file sizes using resolution × colour depth
- **Describe** how analogue sound is converted to digital using sampling
- **Explain** the role of sample rate, bit depth, and sample interval in sound representation
- **Construct expressions** to calculate sound file sizes using sample rate × bit depth × duration
- **Explain** the limitations of binary representation when constrained by available bits
- **Compare** the trade-offs between quality and file size for both images and sound

> **Specification link:** This topic covers Edexcel 1CP2 Paper 1 Topic 2: Data — specifically 2.2.2 (bitmap images), 2.2.3 (sound representation), and 2.2.4 (limitations of binary representation).

Toolkit

These are the key formulas, patterns, and values you need to apply in exam questions. Memorise these — they are essential for calculation questions.

Image file size formulas

Image file size (bits) = width × height × colour depth

Image file size (bytes) = (width × height × colour depth) ÷ 8

Where:

- **Width × height** = total number of pixels (also called resolution)
- **Colour depth** = number of bits used to represent each pixel's colour

Sound file size formulas

Sound file size (bits) = sample rate × bit depth × duration (seconds)

Sound file size (bytes) = (sample rate × bit depth × duration) ÷ 8

For stereo sound, multiply by 2 (for two channels).

Key values to remember

Concept	Common values	What it means
Colour depth	1-bit, 8-bit, 24-bit	1-bit = 2 colours, 8-bit = 256 colours, 24-bit = 16.7 million colours
Sample rate	44,100 Hz (CD quality)	44,100 samples taken per second
Bit depth (sound)	8-bit, 16-bit, 24-bit	Number of bits per sample (more bits = more precision)
Sample interval	1 ÷ sample rate	Time between samples (e.g., 1/44100 ≈ 0.0000227 seconds)

Colour depth and number of colours

Colour depth (bits)	Calculation	Number of colours
1-bit	2^1	2 (black and white)
2-bit	2^2	4
4-bit	2^4	16
8-bit	2^8	256
16-bit	2^{16}	65,536
24-bit	2^{24}	16,777,216 (≈16.7 million, "true colour")

Storage unit conversions

Unit	Abbreviation	Size in bytes	Binary multiplier
Byte	B	1	8 bits
Kibibyte	KiB	1,024	2^{10} bytes
Mebibyte	MiB	1,048,576	2^{20} bytes
Gibibyte	GiB	1,073,741,824	2^{30} bytes

Exam tip: Edexcel uses binary units (kibibyte, mebibyte, gibibyte) rather than decimal units (kilobyte, megabyte, gigabyte). Always use 1024 as the multiplier, not 1000. When the exam says "KB" they typically mean KiB (1024 bytes).

1. Bitmap images

A **bitmap image** is made up of tiny squares called **pixels** (picture elements). Each pixel is a single colour, and the image is formed by arranging thousands or millions of pixels in a grid. The binary data for a bitmap image stores the colour of each pixel.

1.1 Pixels and resolution

The **resolution** of an image refers to the number of pixels it contains, usually expressed as width × height.

- A 1920 × 1080 image has 1920 pixels across and 1080 pixels down
- Total pixels = 1920 × 1080 = 2,073,600 pixels (about 2 megapixels)
- Higher resolution = more pixels = more detail = larger file size

Each pixel needs to be stored in memory, so more pixels means more data to store. When you zoom into a low-resolution image, you can see the individual pixels as coloured squares — this is called **pixelation**.

Understanding resolution

Resolution is often described in different ways:

- **Dimensions:** 1920 × 1080 pixels (width × height)
- **Megapixels:** Total pixels ÷ 1,000,000 (used for cameras)
- **PPI/DPI:** Pixels per inch (for printing) — not tested at GCSE

For calculations, you need the actual pixel dimensions to find total pixels.

1.2 Colour depth

Colour depth is the number of bits used to represent the colour of each pixel. More bits means more possible colours.

- **1-bit colour:** Each pixel is either 0 (black) or 1 (white) — only 2 colours
- **8-bit colour:** Each pixel uses 8 bits — 2^8 = 256 possible colours
- **24-bit colour:** Each pixel uses 24 bits — 2^{24} = 16.7 million colours (true colour)

24-bit colour is often split into three 8-bit values: one for red, one for green, one for blue (RGB). This allows 256 levels of each colour component, which when combined gives 256 × 256 × 256 = 16,777,216 different colours.

> **Why 24-bit is called "true colour":** The human eye can distinguish approximately 10 million colours. With 16.7 million possibilities, 24-bit colour can represent essentially any colour we can see, making images appear photorealistic.

1.3 Calculating image file size

To calculate the file size of a bitmap image, you need to know:

1. The **resolution** (width × height in pixels)
2. The **colour depth** (bits per pixel)

Step-by-step method:

1. Calculate total pixels: width × height
2. Calculate total bits: pixels × colour depth
3. Convert to bytes: divide by 8
4. Convert to larger units if needed: divide by 1024 for KiB, again for MiB

Example calculation

An image is 800 × 600 pixels with 24-bit colour depth. Calculate the file size in mebibytes.

Step 1: Total pixels = 800 × 600 = 480,000 pixels
Step 2: Total bits = 480,000 × 24 = 11,520,000 bits
Step 3: Bytes = 11,520,000 ÷ 8 = 1,440,000 bytes
Step 4: KiB = 1,440,000 ÷ 1024 = 1,406.25 KiB
Step 5: MiB = 1,406.25 ÷ 1024 ≈ 1.37 MiB

1.4 Effect of changing resolution and colour depth

Change	Effect on quality	Effect on file size
Increase resolution	More detail, sharper image	Larger file size
Decrease resolution	Less detail, pixelated	Smaller file size
Increase colour depth	More colours, smoother gradients	Larger file size
Decrease colour depth	Fewer colours, banding	Smaller file size

Exam tip: If you double the resolution in both dimensions (e.g., from 500×500 to 1000×1000), the file size quadruples, not doubles. This is because you're doubling both width AND height.

Worked Example 1 — Describing bitmap image representation

Question: Describe how a bitmap image is represented in binary. [3 marks]

Model answer:
A bitmap image is made up of pixels **(1)**.
Each pixel is stored as a binary value representing its colour **(1)**.
The number of bits used per pixel is called the colour depth **(1)**.

Exam tip: For "describe" questions about image representation, always mention: pixels, binary values for colour, and colour depth. These are the three key concepts examiners look for.

Worked Example 2 — Calculating image file size

Question: An image has a resolution of 1024 × 768 pixels and uses 8-bit colour. Calculate the file size in kibibytes. Show your working. [3 marks]

Model answer:
Step 1: Total pixels = 1024 × 768 = 786,432 pixels **(1)**
Step 2: Total bits = 786,432 × 8 = 6,291,456 bits
Bytes = 6,291,456 ÷ 8 = 786,432 bytes **(1)**
Step 3: KiB = 786,432 ÷ 1024 = 768 KiB **(1)**

Exam tip: Always show your working step by step. Even if you make an arithmetic error, you can still gain method marks. Remember: no calculators are allowed, so practise mental maths with powers of 2.

Worked Example 3 – Effect of changing colour depth

Question: Explain the effect on an image of reducing the colour depth from 24-bit to 8-bit. [2 marks]

Model answer:
The number of available colours would decrease from 16.7 million to 256 **(1)**. This would cause a loss of quality / colour banding / less smooth gradients, but the file size would be smaller **(1)**.

Exam tip: "Explain the effect" questions need both sides: what happens to quality AND what happens to file size. Link cause to effect.

Worked Example 4 – Determining colour depth from file size

Question: An image is 512 × 512 pixels and has a file size of 768 KiB. Calculate the colour depth of the image. [3 marks]

Model answer:
Step 1: Total pixels = 512 × 512 = 262,144 pixels **(1)**
Step 2: File size in bytes = 768 × 1024 = 786,432 bytes
File size in bits = 786,432 × 8 = 6,291,456 bits **(1)**
Step 3: Colour depth = 6,291,456 ÷ 262,144 = 24 bits **(1)**

Exam tip: Rearranging the formula works both ways. If you know the file size and resolution, you can work backwards to find colour depth: colour depth = file size (bits) ÷ number of pixels.

2. Sound representation

Sound in the real world is **analogue** — it's a continuous wave that varies smoothly over time. Computers can only store **digital** data (discrete binary values). To store sound, computers must convert the analogue wave into digital data through a process called **sampling**.

2.1 Analogue vs digital sound

Analogue sound is a continuous wave of varying air pressure. It has infinite possible values at any point in time.

Digital sound is a series of discrete measurements (samples) taken at regular intervals. Each sample is stored as a binary number.

```
ANALOGUE SOUND WAVE (continuous) ●──● ●──● ●──● ● ● ● ● ● ● ● ● ● ● ● ●
──────●──────────────●──────────●──────────────●──────────●──────────────●──────
● ● ● ● ●─● ●─● The wave varies smoothly – infinite possible values at any
point. DIGITAL SAMPLES (discrete measurements) ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
↓ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ▌▌▌
```

Bit depth (also called sample resolution) is the number of bits used to store each sample. It determines how precisely the amplitude can be recorded.

- **8-bit:** 2^8 = 256 possible amplitude levels
- **16-bit:** 2^{16} = 65,536 possible amplitude levels
- **24-bit:** 2^{24} = 16.7 million possible amplitude levels

Higher bit depth means more precise measurements and smoother sound, but larger file sizes. With low bit depth, samples must be rounded to the nearest available level, which introduces **quantisation noise** — a subtle distortion that makes the sound less accurate.

Understanding bit depth practically

Think of bit depth like the number of steps on a staircase:

- **8-bit:** 256 steps — you can get reasonably close to any height, but there are noticeable jumps
- **16-bit:** 65,536 steps — almost any height can be represented with imperceptible rounding

CD quality uses 16-bit because human ears cannot typically detect the difference between 16-bit and higher.

> **Exam tip:** Don't confuse sample rate and bit depth. Sample rate is how OFTEN you measure (frequency). Bit depth is how PRECISELY you measure (accuracy). Both affect quality but in different ways.

2.6 Understanding the sampling process in detail

The conversion from analogue to digital involves several stages that you should understand:

Stage 1: Capture

A microphone converts sound waves (pressure variations in air) into an electrical signal. This signal is still analogue — it varies continuously.

Stage 2: Sampling

At regular intervals (determined by sample rate), the amplitude of the electrical signal is measured. Each measurement is a sample.

Stage 3: Quantisation

Each sample is rounded to the nearest available binary value (determined by bit depth). This introduces a small amount of error.

Stage 4: Encoding

The quantised values are stored as binary numbers in sequence. This is the digital audio data.

2.7 Calculating sound file size

The file size of a sound recording depends on three factors: sample rate, bit depth, and duration.

File size (bits) = sample rate × bit depth × duration (seconds)

For **stereo** sound (two channels — left and right), multiply by 2:

Stereo file size (bits) = sample rate × bit depth × duration × 2

Why stereo doubles the size: Stereo sound has a separate recording for the left speaker and the right speaker. This allows sounds to appear to come from different positions (like in a concert hall). Each channel is a complete audio stream, so the data is doubled.

Example: Calculate the file size of a 3-minute mono (single channel) recording at CD quality (44,100 Hz, 16-bit).
Step 1: Convert duration: 3 minutes = 180 seconds
Step 2: Calculate bits: 44,100 × 16 × 180 = 127,008,000 bits
Step 3: Convert to bytes: 127,008,000 ÷ 8 = 15,876,000 bytes
Step 4: Convert to MiB: 15,876,000 ÷ 1024 ÷ 1024 ≈ 15.14 MiB

2.8 Effect of changing sample rate and bit depth

Change	Effect on quality	Effect on file size
Increase sample rate	Captures more detail, smoother sound	Larger file size
Decrease sample rate	Misses detail, sounds muffled	Smaller file size
Increase bit depth	More precise amplitude, less noise	Larger file size
Decrease bit depth	Less precise, more quantisation noise	Smaller file size

Worked Example 5 – Describing sound sampling

Question: Describe how analogue sound is converted to digital for storage on a computer. [3 marks]

Model answer:

The amplitude/height of the sound wave is measured at regular intervals **(1)**.
Each measurement is called a sample and is stored as a binary value **(1)**.
The number of samples taken per second is called the sample rate **(1)**.

Exam tip: Key terms examiners look for: "amplitude", "sample", "regular intervals", "binary". Use the correct terminology.

Worked Example 6 — Calculating sound file size

Question: A sound file is recorded at 22,050 Hz with 8-bit depth for 2 minutes in mono. Calculate the file size in kibibytes. [3 marks]

Model answer:
Step 1: Duration = 2 × 60 = 120 seconds **(1)**
Step 2: File size = 22,050 × 8 × 120 = 21,168,000 bits
In bytes = 21,168,000 ÷ 8 = 2,646,000 bytes **(1)**
Step 3: In KiB = 2,646,000 ÷ 1024 ≈ 2,583.98 KiB **(1)**

Exam tip: Always convert minutes to seconds first. Watch for mono vs stereo — stereo doubles the file size.

Worked Example 7 — Explaining sample rate effect

Question: Explain why increasing the sample rate improves sound quality but increases file size. [4 marks]

Model answer:
Quality improvement:
A higher sample rate means more samples are taken per second **(1)**.
This captures more detail of the original sound wave, giving a more accurate representation **(1)**.
File size increase:
Each sample requires storage (bit depth worth of bits) **(1)**.
More samples means more data to store, so the file size increases proportionally **(1)**.

Exam tip: For 4-mark "explain" questions, aim for two developed points — each with a statement and expansion/consequence.

Worked Example 8 — Stereo file size calculation

Question: A stereo sound file is recorded at 44,100 Hz with 16-bit depth for 5 minutes. Calculate the file size in mebibytes. [4 marks]

Model answer:
Step 1: Duration = 5 × 60 = 300 seconds **(1)**
Step 2: File size = 44,100 × 16 × 300 × 2 (stereo) = 423,360,000 bits **(1)**
Step 3: In bytes = 423,360,000 ÷ 8 = 52,920,000 bytes **(1)**
Step 4: In MiB = 52,920,000 ÷ 1024 ÷ 1024 ≈ 50.47 MiB **(1)**

Exam tip: Don't forget the ×2 for stereo! This is a common error. Read the question carefully for "mono" or "stereo".

3. Limitations and trade-offs

Binary representation of images and sound always involves trade-offs. The number of available bits constrains what can be stored, creating a balance between quality and file size.

3.1 Limitations of binary representation

When data is constrained by the number of available bits, there are inherent limitations:

Images

- **Limited colours:** With n bits of colour depth, only 2^n different colours can be represented
- **Colour banding:** When colour depth is too low, smooth gradients appear as distinct bands
- **Pixelation:** When resolution is too low, individual pixels become visible
- **Detail loss:** Fine details cannot be captured if there aren't enough pixels

Sound

- **Quantisation error:** Each sample must be rounded to the nearest available value, introducing small errors
- **Limited dynamic range:** With n bits, only 2^n amplitude levels can be stored
- **High frequency loss:** If sample rate is too low, high-pitched sounds are not captured accurately
- **Approximation:** Digital sound is always an approximation of the original analogue wave

Key principle: More bits = better quality = larger file size. There is always a trade-off between quality and storage/transmission requirements.

3.2 Making trade-off decisions

The appropriate settings depend on the purpose:

Scenario	Priority	Recommended settings
Website thumbnail	Small file size	Low resolution, 8-bit colour

Scenario	Priority	Recommended settings
Print photograph	High quality	High resolution, 24-bit colour
Voice message	Small file size	8,000 Hz, 8-bit
Music streaming	High quality	44,100 Hz, 16-bit
Video call	Balance	16,000 Hz, 16-bit

3.3 Metadata

Image and sound files also store **metadata** — additional information about the file that is not the actual image or sound data:

- Image: resolution, colour depth, creation date, camera settings
- Sound: sample rate, bit depth, duration, artist, title

Metadata adds a small amount to the file size beyond the calculated data size. Exam calculations typically ignore metadata.

3.4 Why file sizes in practice differ

Actual file sizes are often different from calculated sizes because:

- **Compression:** Most image and sound formats use compression to reduce file size
- **Metadata:** Extra information is stored alongside the media data
- **File headers:** Technical information about how to read the file

> **Exam tip:** Exam questions ask you to calculate the **uncompressed** file size. The formula gives the raw data size before compression is applied.

Worked Example 9 – Explaining limitations

Question: Explain two limitations of representing images in binary with a low colour depth. [4 marks]

Model answer:
Limitation 1 — Fewer colours available:
A low colour depth means fewer bits per pixel, so fewer colours can be represented **(1)**.
For example, 4-bit colour only allows 16 colours compared to 16.7 million with 24-bit **(1)**.
Limitation 2 — Colour banding:
Smooth gradients in the original image cannot be accurately represented **(1)**.
This results in visible bands/steps between colours rather than smooth transitions **(1)**.

> **Exam tip:** For limitation questions, name the limitation clearly, then explain why it occurs or give an example to illustrate.

Worked Example 10 – Comparing settings

Question: A music streaming service offers two quality settings: Standard (22,050 Hz, 16-bit, stereo) and High (44,100 Hz, 16-bit, stereo). Compare the file sizes of a 4-minute song at each setting. [4 marks]

Model answer:
Standard quality:
22,050 × 16 × 240 × 2 = 169,344,000 bits = 21,168,000 bytes ≈ 20.19 MiB **(1)**
High quality:
44,100 × 16 × 240 × 2 = 338,688,000 bits = 42,336,000 bytes ≈ 40.37 MiB **(1)**
Comparison:
The high quality file is exactly twice the size of the standard quality file **(1)**.
This is because the sample rate has doubled while bit depth and channels remain the same **(1)**.

Exam tip: For comparison questions, calculate both values, then explicitly state the relationship or difference between them.

Worked Example 11 – Trade-off scenario

Question: A photographer needs to email a large image. Describe two ways they could reduce the file size while maintaining acceptable quality. [4 marks]

Model answer:
Method 1 — Reduce resolution:
Decrease the number of pixels in the image **(1)**.
This reduces the total amount of data while still maintaining colour quality if the image doesn't need to be viewed at full size **(1)**.
Method 2 — Reduce colour depth:
Reduce from 24-bit to 16-bit colour **(1)**.
This halves the amount of data per pixel while still providing enough colours for most photographs **(1)**.
Also acceptable: Use compression (though this is covered in a separate topic)

Exam tip: Scenario questions want practical answers. Link each method to how it reduces file size and acknowledge any quality impact.

Worked Example 12 – Extended calculation

Question: A video game stores background music as uncompressed audio. The game has 15 music tracks, each 4 minutes long, recorded at 44,100 Hz, 16-bit,

stereo. Calculate the total storage required for all music in gibibytes. [5 marks]

Model answer:
Step 1: Duration per track = 4 × 60 = 240 seconds **(1)**
Step 2: Size per track = 44,100 × 16 × 240 × 2 = 338,688,000 bits
In bytes = 338,688,000 ÷ 8 = 42,336,000 bytes **(1)**
Step 3: Total for 15 tracks = 42,336,000 × 15 = 635,040,000 bytes **(1)**
Step 4: In KiB = 635,040,000 ÷ 1024 = 620,156.25 KiB
In MiB = 620,156.25 ÷ 1024 ≈ 605.62 MiB **(1)**
Step 5: In GiB = 605.62 ÷ 1024 ≈ 0.59 GiB **(1)**

Exam tip: For extended calculations, show every step clearly. Breaking it down helps you avoid errors and ensures you get method marks even if the final answer is slightly off.

Worked examples summary

The table below summarises the 12 worked examples covered in this topic, organised by section and skill type.

#	Topic	Question type	Marks	Key skill
1	Images	Describe representation	3	Pixels, binary, colour depth
2	Images	Calculate file size	3	Apply formula, convert units
3	Images	Explain effect	2	Link change to quality/size
4	Images	Reverse calculation	3	Find colour depth from size
5	Sound	Describe sampling	3	Amplitude, sample, sample rate
6	Sound	Calculate file size	3	Apply formula, convert units
7	Sound	Explain sample rate	4	Quality and size effects
8	Sound	Stereo calculation	4	Remember ×2 for stereo
9	Limitations	Explain limitations	4	Name and explain effects
10	Trade-offs	Compare settings	4	Calculate both, state relationship
11	Trade-offs	Reduce file size	4	Practical methods
12	Extended	Multi-step calculation	5	Multiple files, unit conversion

Quick reference

Key terms and definitions

Term	Definition (mark-scheme style)
Pixel	The smallest addressable element of a bitmap image; a single coloured dot
Resolution	The number of pixels in an image, typically expressed as width × height
Colour depth	The number of bits used to represent the colour of each pixel
Bitmap	An image made up of a grid of pixels, each storing a colour value
Sample	A single measurement of sound wave amplitude at a point in time
Sample rate	The number of samples taken per second, measured in Hertz (Hz)
Bit depth (sound)	The number of bits used to store each sample; determines precision of amplitude
Sample interval	The time between each sample; equals 1 ÷ sample rate
Amplitude	The height/loudness of a sound wave at a given point
Analogue	Data that varies continuously with infinite possible values
Digital	Data represented as discrete binary values (0s and 1s)
Quantisation	Rounding analogue values to the nearest available digital value

Formulas at a glance

What to calculate	Formula	Example
Number of colours	$2^{\text{colour depth}}$	2^8 = 256 colours
Image size (bits)	width × height × colour depth	800 × 600 × 24 = 11,520,000 bits
Image size (bytes)	(width × height × colour depth) ÷ 8	11,520,000 ÷ 8 = 1,440,000 bytes
Sound size (bits)	sample rate × bit depth × duration	44,100 × 16 × 60 = 42,336,000 bits
Stereo sound (bits)	sample rate × bit depth × duration × 2	44,100 × 16 × 60 × 2 = 84,672,000 bits

What to calculate	Formula	Example
Sample interval	1 ÷ sample rate	1 ÷ 44,100 ≈ 0.0000227 s
Bits to bytes	bits ÷ 8	8,000,000 ÷ 8 = 1,000,000 bytes
Bytes to KiB	bytes ÷ 1024	1,048,576 ÷ 1024 = 1024 KiB
KiB to MiB	KiB ÷ 1024	1024 ÷ 1024 = 1 MiB

Common values to remember

Value	Meaning
44,100 Hz	CD quality sample rate
16-bit	CD quality bit depth
24-bit colour	True colour (16.7 million colours)
8 bits = 1 byte	Basic conversion
1024 bytes = 1 KiB	Binary unit conversion
1024 KiB = 1 MiB	Binary unit conversion

Typical pitfalls

Pitfall 1 — Forgetting to convert units.
Wrong: Giving the answer in bits when the question asks for bytes or KiB.
Right: Always check what unit the question asks for, then divide by 8 for bytes, by 1024 for KiB, etc.

Pitfall 2 — Using 1000 instead of 1024.
Wrong: Converting 1,000,000 bytes to "1000 KB".
Right: Edexcel uses binary units: 1,000,000 ÷ 1024 = 976.56 KiB.

Pitfall 3 — Forgetting stereo doubles the size.
Wrong: Calculating stereo sound the same as mono.
Right: Stereo has two channels, so multiply by 2. Read the question carefully for "mono" or "stereo".

Pitfall 4 — Confusing sample rate and bit depth.
Wrong: Saying "higher sample rate means more precise amplitude".

Right: Sample rate = how often samples are taken (frequency). Bit depth = how precisely each sample is measured (precision).

Pitfall 5 — Mixing up resolution and colour depth.
Wrong: "Increasing resolution adds more colours".
Right: Resolution = number of pixels. Colour depth = bits per pixel (number of colours). They are independent settings.

Pitfall 6 — Forgetting to convert minutes to seconds.
Wrong: Using "3" directly when the duration is 3 minutes.
Right: Always convert: 3 minutes = 180 seconds. Sample rate is per second.

Pitfall 7 — Incomplete descriptions of sampling.
Wrong: "Sound is converted to binary".
Right: Include key details: amplitude is measured, at regular intervals, stored as binary values. Use technical terms.

Pitfall 8 — Not linking effect to cause.
Wrong: "Higher resolution means larger file size" (no explanation).
Right: "Higher resolution means more pixels, and each pixel needs to be stored, so the file size increases."

Pitfall 9 — Confusing bits and bytes in colour depth.
Wrong: "24-byte colour depth".
Right: Colour depth is always in bits (1-bit, 8-bit, 24-bit). Be careful with terminology.

Pitfall 10 — Not showing working in calculations.
Wrong: Just writing the final answer.
Right: Show each step: pixels × colour depth = bits, ÷8 = bytes, ÷1024 = KiB. Method marks are available.

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